

Smaran Vallabhaneni

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EDUCATION

Olin College of Engineering <i>Bachelor of Science, Engineering in Computing (GPA: 4.0)</i> – Relevant Courses: Modeling & Simulation, Linear Algebra, Data Science, Multi-Variable Calculus, Software Design (Python), Kernel Development (C), Products & Markets, Introduction to Sensors, Instrumentation & Measurement	Needham, MA <i>Aug. 2025 – May 2029</i>
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SKILLS

Programming Languages: Java, C/C++, Python, pandas, HTML, CSS, JavaScript, Dart, L ^A T _E X, MySQL, NoSQL, C#, R, PBASIC, Bash, Zsh, PowerShell, Swift, MATLAB
Operating Systems / SBCs: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based), Raspberry Pi, Arduino
Tools: Blender, Fusion360, AutoCAD, Autodesk, SolidWorks, Unity, Figma, Adobe Creative Suite, Excel, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Overleaf, Node.js, GPL Cver, Simulink, MATLAB, Tableau Desktop, Docker, MissionPlanner, ArduPilot, GitHub Copilot, Google Gemini, Claude Code
Frameworks: LLMs, RAG Systems, Django, PyTorch, NumPy, Tkinter, Scikit-Learn, Kivy, Flutter, SwiftUI, React, AngularJS, .NET, Ollama, System Design, Metal API, ROS/MAVROS, OpenCV, MediaPipe, Blender Python API, OpenStreetMap
Other: UI/UX, CI/CD Pipelines, ESP32, Computer Vision, Internet of Things, AWS, GCP, Azure, Dev Ops, HuggingFace, NLP, Cryptography, Encryption, Databases, Database Management, Statistical Analysis, Version Control, Embedded Systems, Algorithm Development, Data Structures, Object-Oriented Programming, API Integration, Testing and Debugging
Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail
Languages: English, Spanish, French, Chinese, German (Limited Working), Telugu

PROJECTS

CLOS (Command Line Operating System) <i>C++, Java, HTML, CSS, Bash</i> – Building a cross-platform centralized terminal and graphical math tool with equation solving, graphing, symbolic calculus, and trigonometry. – Engineered a custom database engine, AES encryption module, symbolic math parser, and a 3D rendering library from scratch in Java; achieved consistent 60 FPS graphic rendering and executed complex multi-variable calculus computations under 50 ms.	Apr. 2023 – Present
ClassicTunes <i>Swift, SwiftUI, Metal API</i> – Developed an open-source modern remake of iTunes 7–10 for Apple Silicon macOS, faithfully recreating CoverFlow, MiniPlayer, the iTunes Store, iPod Syncing, and Smart/Genius Playlists. – Leveraged Apple’s Metal API to build a custom GPU-accelerated CoverFlow engine, matching the performance of the native Apple Music client with a fluid 120 FPS render rate across 150+ albums on Apple Silicon.	Oct. 2025 – Present
Whoa-Scope <i>Python, Kivy</i> – Co-developed an open-source oscilloscope fork to resolve compatibility issues and desktop UX flaws, gaining organic adoption from 50% of the class (40 students) and faculty recognition. – Profiled and refactored memory allocation, slashing RAM consumption by 200+ MB to maximize stability and prevent lag on lab computers.	Feb. 2026 – Present
Specter <i>Python, Linux, LLMs</i> – Created a local-first, privacy oriented AI screen agent for Fedora Wayland to perform tasks efficiently with simple prompts. – Locally ran with a response time of 3 seconds per action, with extreme fidelity and understanding of on-screen content on a capable GPU.	May 2026
3D Gesture Tracking <i>Blender, Python, OpenCV</i> – Engineered a real-time computer vision add-on published on Blender Addons, enabling mouse-free viewport manipulation (pan, zoom, rotate) via hand gesture tracking.	Apr. 2026
OSM2Mesh <i>Python, OpenStreetMap</i> – Built an easy-to-use map to 3D model city generator that produces high-quality, game-ready city models with textures within 20 seconds. Designed for all standard 3D and game engines with extensive customizability and an easy-to-use cross-platform UI.	Jul. 2026

EXPERIENCE

Olin College AERO Project Team <i>Head of Ground Station Development — Software Engineer</i> – Leading software development for an autonomous fixed-wing aircraft competing in the AUVSI SUAS competition, targeting precision water delivery to GPS-specified targets using image recognition. – Architected ground station software integrating ArduPilot, MissionPlanner, and ROS/MAVROS systems to bridge an on-board NVIDIA Jetson computer with ground controls. – Engineered a robust fly-by-wire telemetry pipeline, achieving a consistent sub-100ms round-trip latency for real-time sensor monitoring and safety interlocks.	Needham, MA <i>Sep. 2025 – Present</i>
Geeky Insights <i>Full-Stack Android App Developer — Internship</i> – Designed end-to-end UI/UX of a customizable audio alarm clock app using Figma and Adobe Creative Suite, prioritizing accessibility and intuitive design for a diverse user base. – Engineered an end-to-end frontend/backend architecture using Flutter, Dart, and Java, optimizing the compilation pipeline into a lightweight, bug-free 15 MB production package. – Optimized application memory layout to support low-resource devices running Android 9+, reducing RAM consumption to match the baseline footprint of Android’s native system clock.	Remote <i>Apr. 2024 – Apr. 2025</i>

ACTIVITIES

Olin ICPC <i>Competitive Programmer — Team Member</i> – Solving high-difficulty algorithmic problems under competitive time constraints in bi-weekly ICPC-style contests. – Applying advanced techniques including segment/Fenwick trees, dynamic programming, graph algorithms, string algorithms, network flow, and computational geometry.	Needham, MA <i>Jan. 2026 – Present</i>
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